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Concept Review

Load Cell

Why Use a Load Cell?

A load cell is a transducer that converts a mechanical force into an electrical signal. They are used for the accurate, reliable, and repeatable measurements of load in practical applications such as industrial weighing systems, material testing machines, and structural monitoring. By converting mechanical force into an electrical output, load cells enable easy data collection, real-time monitoring, and integration with automated control systems.

Measuring Principle

A load cell converts a mechanical force or weight into an electrical signal through the measurement of strain in a structural element. When a force is applied to a load cell, it causes a small deformation in the material. Strain is a measure of deformation in a solid body due to an applied force. Figure 1 shows a specimen subjected to an axial tensile stress, σ.

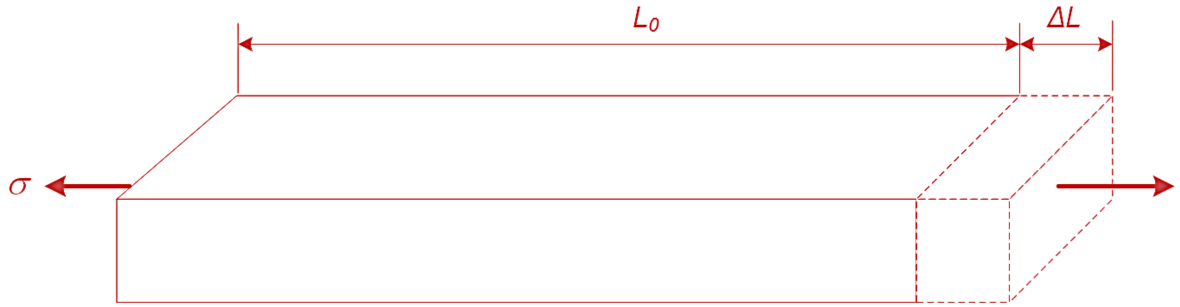

Figure 1: Change in length due to an axial tensile force

This stress causes a change in the original length of the bar from L_0 to $L_0 + \Delta L$. We define strain, ε, using the following equation:

$$\varepsilon = \frac{\Delta L}{L_0}$$

where L_0 is the original length and ΔL is the change in length due to the applied force. Strain is dimensionless and expressed as a percentage (%) or in mm/mm. However, since strain values are typically very small, strain is expressed in micro-strain (με) by multiplying strain by 10^6.

Figure 2 illustrates the relationship between stress and strain when applied to a solid body. As shown in the graph, as stress increases, the solid body undergoes various deformation stages. In the elastic region, the body does not experience permanent physical change. In this region, the stress-strain relationship exhibits a linear relationship, which translates to the functional range of the load cell.

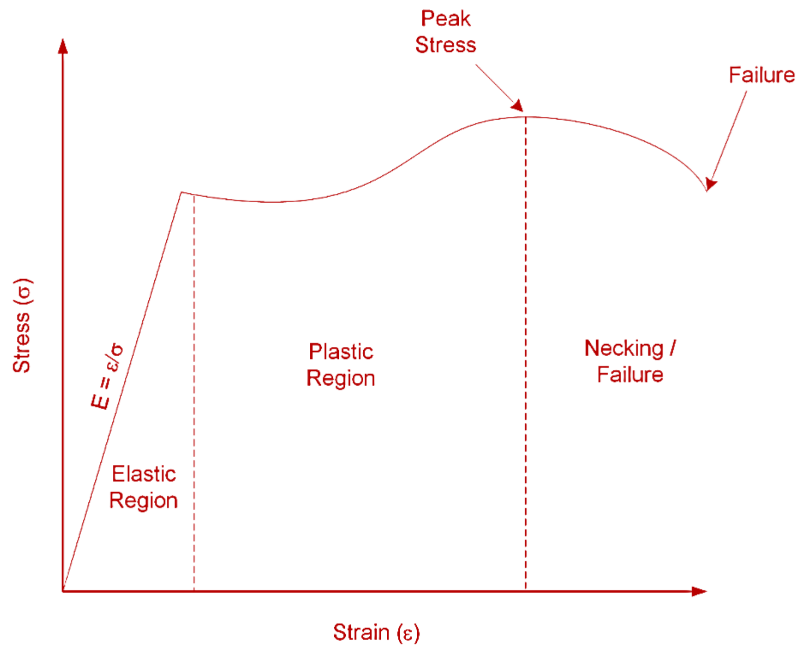

Figure 2: Stress-strain curve

Strain Gage

Load cells often consist of strain gages bonded to a material within the load cell. A strain gage is a sensor used for measuring strain in solid bodies. As shown in Figure 3, it is constructed from a fine metallic foil element formed into a grid pattern and mounted on a thin backing called a carrier. Strain gages are commonly bonded to test specimens using cyanoacrylate-based adhesives or two-part epoxies. When an appropriate bond between the gage and specimen is established, any deformation in the specimen is transferred to the gage. This causes the resistance of the strain gage to change. When a strain gage is under tension its resistance increases, while under compression its resistance decreases.

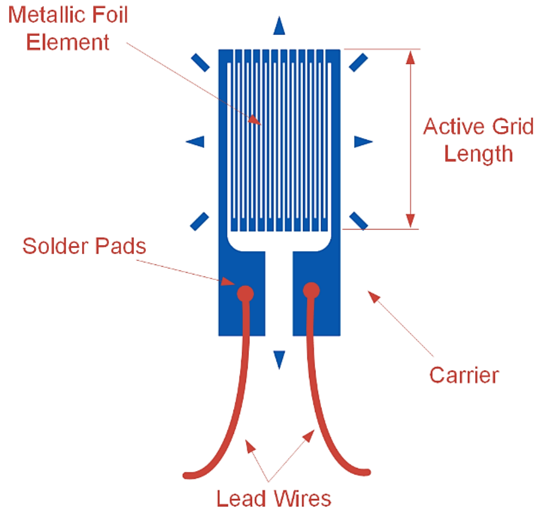

Figure 3: Schematic diagram of a strain gage

Strain gages vary in shape, orientation, and number depending on the type of strain being measured. The output of a strain gage is measured by connecting its lead wires to a dedicated signal conditioning circuit and DAQ, or dedicated strain measuring instrument. The relationship between the relative change in resistance and the strain is characterized by the **gage factor (GF)**:

$$GF = \frac{\frac{\Delta R}{R_G}}{\varepsilon}$$

where ΔR is the change in resistance when the strain gage is deformed, R_G is the nominal resistance of the gage, and ε is the induced strain. A typical gage factor value is approximately 2. For example, $GF = 2$ means if 1% strain is induced in a specimen, the gage's relative resistance will change by 2%, or 20,000 με.

Wheatstone Bridge Circuits

To convert the change in resistance into a measurable voltage, strain gages are typically arranged in a Wheatstone bridge circuit. As the load cell deforms, the bridge becomes unbalanced, producing an output voltage proportional to the applied load. This small voltage signal is then amplified and conditioned for measurement and data acquisition.

Wheatstone bridge circuit offers several advantages over voltage dividing circuits, which are typically used for measuring the output of resistive sensors. One benefit is that a Wheatstone bridge allows for higher measurement sensitivity and lower measurement error. Another advantage is that it removes large, fixed voltage drops which are present in a typical voltage dividing circuit. A Wheatstone bridge consists of four resistive arms, R_1, R_2, R_3, and R_4, and is excited by a voltage source, V_{Ex}, as shown in Figure 4. The output voltage, V_O, is measured between the midpoints of the two resistor pairs.

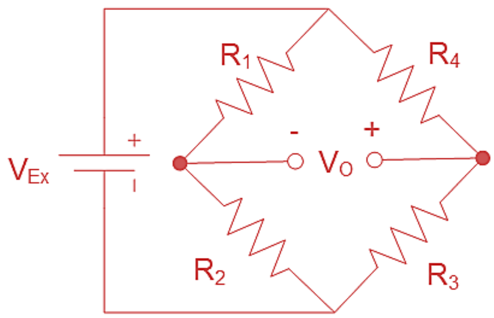

Figure 4: A Wheatstone bridge circuit for measuring the output of resistive sensors

The output voltage is governed by the following equation:

$$V_O = \left(\frac{R_3}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}\right) V_{Ex}$$

When the bridge is balanced, the ratios of resistances satisfy $R_1/R_2 = R_4/R_3$, producing an output voltage $V_O = 0$. However, when the resistance of one or more arms changes due to strain in the load cell, the bridge becomes unbalanced and produces an output voltage proportional to the applied strain (and therefore to the applied force or load). Because the output is typically in the microvolt range, amplification is required before data acquisition.

Three common Wheatstone bridge configurations are used in load cell and strain measurement applications: quarter-bridge, half-bridge, and full-bridge configuration.

Quarter-Bridge Configuration

The quarter-bridge setup (Figure 5) uses one active strain gage and three fixed precision resistors. It is the simplest configuration but provides the lowest sensitivity. When a load is applied, the strain in the gage changes its resistance, creating a small output voltage.

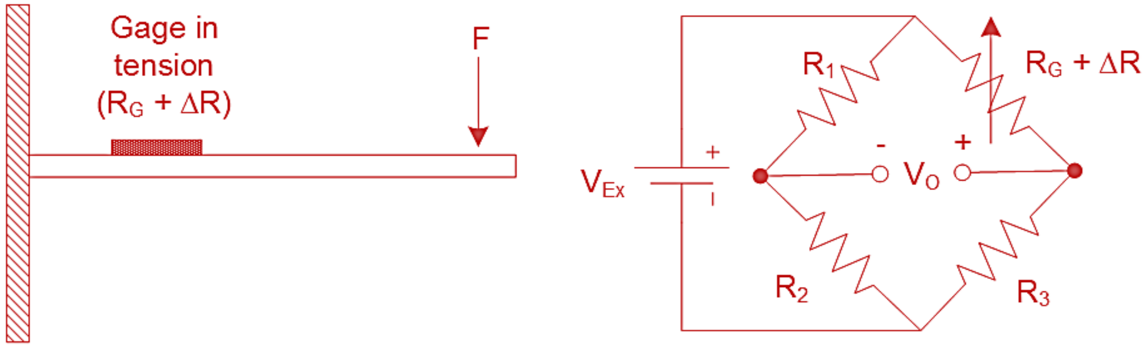

Figure 5: Quarter-bridge configuration for measuring strain in a cantilever beam

When the four resistors have the same initial resistance, $R_1 = R_2 = R_3 = R_G$, the output voltage, V_O, of the quarter-bridge circuit can be expressed in terms of V_{Ex}, GF, and the measured strain:

$$\frac{V_O}{V_{Ex}} = -\frac{GF \cdot \varepsilon}{4}\left(\frac{1}{1 + GF \cdot \frac{\varepsilon}{2}}\right)$$

Half-Bridge Configuration

A half-bridge circuit (Figure 6) employs two strain gages and two fixed resistors. It offers the benefit of twice the sensitivity of a quarter-bridge configuration. The two gages are often mounted on opposite sides of a beam. When a bending force is applied to the beam, it causes tension in one of the gages while the other gage compresses.

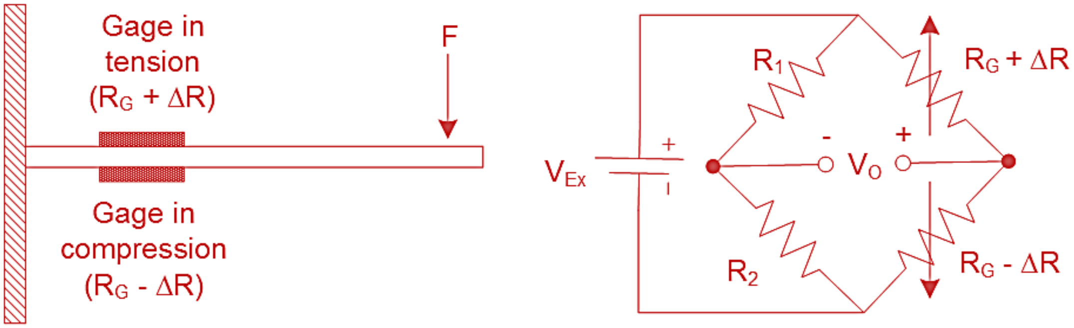

Figure 6: Half-bridge configuration for measuring strain in a cantilever beam

When $R_1 = R_2 = R_G$, the output voltage, V_O, of the half-bridge circuit can be expressed in terms of V_{Ex}, GF, and the measured strain:

$$\frac{V_O}{V_{Ex}} = -\frac{GF \cdot \varepsilon}{2}$$

Full-Bridge Configuration

In the full-bridge configuration (Figure 7), all four arms of the bridge are active strain gages. This setup provides the highest measurement sensitivity and best compensation for temperature and lead-wire resistance.

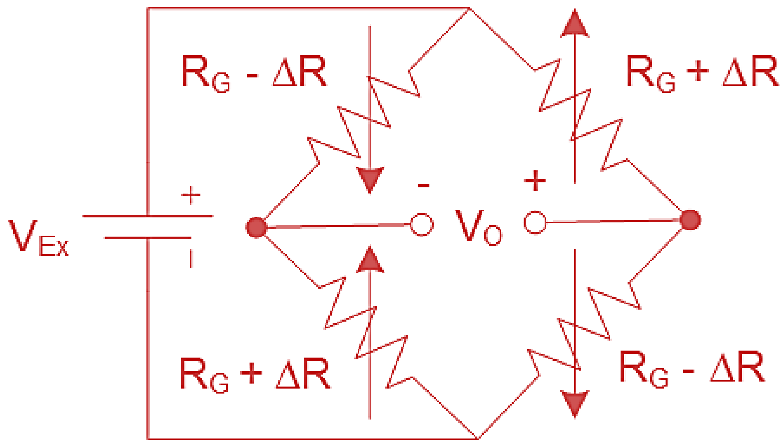

Figure 7: Full-bridge configuration for measuring strain

A full-bridge configuration produces twice the sensitivity of a half-bridge configuration and four times the sensitivity of a quarter-bridge configuration. The corresponding relationship can be expressed as:

$$\frac{V_O}{V_{Ex}} = -GF \cdot \varepsilon$$

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